Polar Bayesian decoding: tractable approximations

From effective finite-alphabet receivers to feasible computation

29 October 2026 — Report 11; continuing algebraic investigation

Research aim. The approximations developed so far improve decoding performance while limiting the search over possible transmissions. The resulting receivers nevertheless retain computations over the full message alphabet. The next objective is to make decoding feasible as message length and activity grow, allowing an explicit loss of accuracy where it buys useful computational savings. We pursue this through nonlearned, algebraically motivated approximations, using the exact observations in Report 7 and the approximation results in Report 9 as foundations. The fixed encoder and PUPE objective remain unchanged.

1 Keep the target; identify the computation to remove

Write $b = B_c$, $d = N_c$, $P = 2^{B_p}$ and $M = P2^b = 2^B$. With the balanced encoder of Report 8,

$$\phi(p, v) = S_p D_p (-1)^{Lv}, \qquad D_p = \operatorname{diag}(\sqrt{e_p}), \qquad y = \sum_{k=1}^{K} \phi(w_k) + z, \quad z \sim \mathcal{N}(0, \sigma^2 I_n). \tag{1}$$

The receiver knows the encoding rule, K and $\sigma^2 > 0$. Messages and their patterns are unknown. Every waveform has unit energy. IID uniform messages give the count posterior

$$\pi_y(a) = \frac{1}{Z_y} \frac{K!}{M^K \prod_w a_w!} \exp\left[-\frac{\|y - \sum_w a_w \phi(w)\|^2}{2\sigma^2} \right], \qquad \sum_w a_w = K. \tag{2}$$

For an output count vector c totaling K, the conditional PUPE is

$$\mathcal{R}_y(c) = 1 - \frac{1}{K} \sum_w \sum_{j=1}^{c_w} T_{w,j}, \qquad T_{w,j} = \mathbb{P}(a_w \geq j \mid y). \tag{3}$$

The Bayes action repeatedly selects the largest available next tail. Approximations change how these tails are obtained; a likely configuration or a small signal residual alone is a different decision rule. Distinct-message sampling is a separate conditioned source, which still allows shared patterns.

There are three costs to separate. Payload processing should avoid enumerating 2^b words. The fixed placement/energy bank already contains Pd entries, so retaining it costs $\Theta(Pd)$. Finally, the number of candidate configurations, samples and residual searches can grow with K. Removing one exponential sum does not bound the others.

Current multiset-beam scoring costs roughly $O(KWnM)$ for width W; sparse columns can replace n by d but leave M. The B12/B14 experiments retain $b = 6, d = 32, n = 256$ and increase P from 64 to 256. They therefore identify useful inference mechanisms without testing longer polar payloads. A payload scaling study must increase b, d at a declared rate and also specify P, n, K and noise/energy scaling.

2 Use the exact conditional polar problem

Given a hypothesized configuration of the other $K - 1$ slots, form $r = y - \sum_{j \neq k} \phi(w_j)$. Unit candidate energy gives

$$f_r(p, v) = e^{s_p(v)}, \qquad s_p(v) = \frac{r^T \phi(p, v)}{\sigma^2} = \sum_{i=1}^{d} h_{p,i} (-1)^{\ell_i^T v}, \qquad h_{p,i} = \frac{\sqrt{e_{p,i}} r_{r(p,i)}}{\sigma^2}. \tag{4}$$

Here ℓ_i^T is row i of L. These are exact conditional weights, with polar LLRs $2h_{p,i}$. The outside messages are receiver hypotheses, not known true transmissions.

2.1 A polar list is a proposal, not a partition

SCL can return a bounded list without enumerating payloads. It does not supply $Z_p = \sum_v e^{s_p(v)}$ or an exact conditional sample. Its retained path metric is

$$m_p(v) = \sum_i \log(2\cosh h_{p,i}) - s_p(v).$$

The first term varies with the pattern. Cross-pattern comparisons must restore it or directly rescore the physical waveforms. An empty CRC-valid list does not establish pattern absence.

For a deduplicated list $C(r)$ and declared guide scores $\widetilde{s}$, an available normalized proposal is

$$q_C(w \mid r) = \frac{\mathbf{1}\{w \in C(r)\}e^{\widetilde{s}(w)}}{\sum_{u\in C(r)} e^{\widetilde{s}(u)}}, \qquad q(w \mid r) = (1-\alpha)q_C(w \mid r) + \alpha/M, \tag{5}$$

where $0 < \alpha < 1$; use the uniform law if the list is empty. Random message bits supply the uniform draw without an alphabet scan. A single-slot Metropolis–Hastings (MH) move accepts $u \to w$ with

$$A(u, w \mid r) = \min\{1, e^{s_r(w)-s_r(u)}q(u \mid r)/q(w \mid r)\}. \tag{6}$$

The outside residual is identical in both directions. A sequential pair proposal needs its actual reverse contexts as well. These constructions already exist in the earlier revisable-sampling prototype. Full support ensures a valid target, but a missed word may still require order M/α draws to discover.

2.2 Residual uncertainty changes the guide

When h users remain, including the next proposed word, Report 9's Gaussian continuation replaces the unresolved sum by its mean and covariance. With one-message moments μ, Σ_x, put

$$u = r - (h-1)\mu, \qquad \Sigma_h = \sigma^2 I + (h-1)\Sigma_x, \qquad \widetilde{s}(x) = u^T\Sigma_h^{-1}x - \tfrac{1}{2}x^T\Sigma_h^{-1}x.$$

This is an approximation to the remaining-user convolution, not a change in physical noise. For scalar Σ_h, the quadratic term is common and ordinary polar LLRs apply. A diagonal covariance adds a pattern-dependent constant. A dense covariance introduces additional parity terms:

$$\begin{aligned}\widetilde{s}_p(v) = &\sum_i a_i(S_p^T\Sigma_h^{-1}u)_i\chi_{\ell_i}(v) \\ &- \tfrac{1}{2}\sum_{i,j} a_i a_j(S_p^T\Sigma_h^{-1}S_p)_{ij}\chi_{\ell_i+\ell_j}(v),\end{aligned} \tag{7}$$

where $a_i = \sqrt{e_{p,i}}$ and $\chi_g(v) = (-1)^{g^T v}$. Thus a dense guide is generally not an ordinary d-symbol SCL input. Dropping its extra terms is a further approximation. Exact physical weighting can correct a guide's target bias, but cannot guarantee useful sampling or fast mixing.

3 Build a normalized reference from independent parities

Report 9's character algebra gives another tractable inner operation. For a fixed pattern and affine payload restriction $v = v_0 + Rz$, $z \in \mathrm{GF}(2)^q$, group identical restricted rows:

$$s(z) = H_0 + \sum_{g\in\mathcal{G}} H_g\chi_g(z), \qquad g \neq 0. \tag{8}$$

Grouping is exact. The physical conditional has at most d groups; the dense Gaussian guide may have $O(d^2)$. Choose an independent row set F, of rank r_F, and split the score into

$$\begin{aligned} s_F(z) &= H_0 + \sum_{g\in F} H_g\chi_g(z), & D_F(z) &= \sum_{g\notin F} H_g\chi_g(z), \\ R_F &= \sum_{g\notin F} |H_g|, & Z_F &= 2^{q-r_F}e^{H_0}\prod_{g\in F} 2\cosh H_g. \end{aligned} \tag{9}$$

Independent parity assignments have equal multiplicity 2^{q-r_F}. Consequently $q_F(z) = e^{s_F(z)}/Z_F$ is normalized. Sample each retained parity independently with $\mathbb{P}(g^T z = 0) = e^{H_g}/(2\cosh H_g)$, then solve the binary system and choose its free bits uniformly. No payload enumeration or unknown normalizer is required.

Used alone, q_F ignores the dependent terms. Alternatively it is a reference for exact correction. Since $|D_F| \le R_F$,

$$L_F := e^{-R_F} Z_F \le Z := \sum_z e^{s(z)} \le U_F := e^{R_F} Z_F, \qquad e^{s(z)} \le U_F q_F(z). \tag{10}$$

We call the last pointwise upper bound an *envelope*. It bounds both a partition and the weight of each possible word.

3.1 Choose the basis and combine unknown patterns

Greedy selection in decreasing $|H_g|$, keeping a row only if independent, minimizes R_F. It also minimizes U_F within this family: retaining a row reduces the log upper bound by $|H_g| - \log\cosh H_g$, which increases with $|H_g|$. The independent-set exchange property gives the same greedy ordering for either weight. This is optimality of this bound construction, not of the decoder.

For unconditioned patterns set $q = b$ and $U_{\text{all}} = \sum_p U_{F,p}$. Propose a pattern and payload by

$$Q(p, z) = \frac{U_{F,p}}{U_{\text{all}}} q_{F,p}(z), \qquad a(p, z) = e^{D_{F,p}(z) - R_{F,p}}. \tag{11}$$

Then $Qa = e^{s_p(z)}/U_{\text{all}}$. Rejection sampling therefore returns an exact one-slot conditional, with acceptance $Z_{\text{all}}/U_{\text{all}}$ and mean attempts $U_{\text{all}}/Z_{\text{all}}$. The unknown partition controls runtime even though it is unnecessary to execute a draw.

The same bound certifies a list's conditional omitted mass. If its exact weight is Z_C, then

$$\mathbb{P}(w \notin C \mid r) \le 1 - Z_C/U_{\text{all}}. \tag{12}$$

This concerns one specified outside configuration. It does not certify that a multiuser search has found the right configurations. For a distinct-message source, reject prohibited full words and use the allowed partition.

3.2 A finite update need not wait for rejection to succeed

Cap rejection at L_a attempts and keep the current word on failure. For a fixed outside context this gives

$$\beta(r)\pi_r + (1 - \beta(r))I, \qquad \beta(r) = 1 - (1 - Z_{\text{all}}/U_{\text{all}})^{L_a},$$

where π_r is the exact conditional. This kernel is invariant because its success probability does not depend on the current slot value. Using Q in Eq. (6) is another bounded-cost update. Neither result establishes rapid movement of the full K-user chain.

3.3 Dependent parities can make the envelope poor

Suppose the score contains $g, h, g + h$ with coefficients $(a, a, -a)$, $a > 0$. Writing $x = \chi_g$, $u = \chi_h$ gives $s = a(x + u - xu)$: three assignments score a and one scores $-3a$. Any two-character reference has acceptance

$$\mathcal{A}(a) = \frac{3e^a + e^{-3a}}{e^a (2\cosh a)^2} \sim 3e^{-2a}. \tag{13}$$

The preferred signs conflict, so separate factor bounds cannot be attained together. The selected $b = 6, d = 32$, CRC8 generator contains this triangle among its 28 distinct nonzero rows. Positive amplitudes and distinct occupied resources allow a residual to realize these coefficients and zero others. This is a realizable conditional obstruction, not a measurement of typical residuals.

4 Refine selected states without discarding the rest

A loose envelope need not be replaced everywhere. Suppose $E(w) = UQ(w) \geq f(w) > 0$, with an available normalized proposal Q. Evaluate exact weights on a finite set A and define

$$Z_A = \sum_{w \in A} f(w), \quad \tau_A = Q(A^c), \quad E_A(w) = \begin{cases} f(w), & w \in A, \\ E(w), & w \notin A, \end{cases} \quad U_A = Z_A + U\tau_A. \tag{14}$$

Then $f \leq E_A \leq E$ and $Z \leq U_A \leq U$. Exploring w reduces the upper bound by $E(w) - f(w)$. A useful state to refine is therefore one whose weight the reference overestimates; it need not be a particularly likely transmitted message.

The unknown outside mass is at most $U\tau_A$, giving

$$\mathbb{P}(w \notin A \mid r, p) \leq \frac{U\tau_A}{Z_A + U\tau_A}. \tag{15}$$

If a lower envelope $LQ \leq f$ is also available, the partition interval becomes $Z_A + L\tau_A \leq Z \leq Z_A + U\tau_A$.

4.1 The refined envelope is an explicit full-space proposal

With probability Z_A/U_A, draw an explored word proportionally to its exact weight and accept. Otherwise draw from Q conditioned on A^c and accept with probability $f(w)/E(w)$. The proposal is E_A/U_A, its accepted density is f/U_A, and its acceptance is Z/U_A. Mean attempts decrease after refinement, but the work of constructing A must still be recovered. Across patterns, select p proportionally to its refined $U_{A,p}$.

The same proposal can be used without rejection for importance sampling. With $\pi = f/Z$,

$$\sup_w \frac{\pi(w)}{E_A(w)/U_A} \leq \frac{U_A}{Z}, \qquad 1 + \chi^2(\pi \| E_A/U_A) \leq \frac{U_A}{Z}. \tag{16}$$

The second inequality follows by bounding one factor of the squared density ratio and averaging the other. Refinement improves this variance bound; it does not prove that its preparation is worthwhile.

4.2 Sample outside an arbitrary list directly

Repeatedly drawing Q until it leaves A would cost $1/\tau_A$ attempts. The parity reference avoids this problem. An invertible binary coordinate change gives independent retained parities followed by uniform free coordinates. Transform the explored words into these coordinates and build a prefix tree.

Let $m(\alpha)$ be the conditional probability of avoiding explored leaves below prefix α. Use $m = 1$ where no explored descendant exists, $m = 0$ at an explored leaf, and otherwise

$$m(\alpha) = (1 - p_i)m(\alpha 0) + p_i m(\alpha 1). \tag{17}$$

Here p_i is the product law's probability of bit one. Choose the next bit c with probability $\mathbb{P}(X_i = c)m(\alpha c)/m(\alpha)$. The factors telescope to $Q(w)/\tau_A$ outside the list. This is a direct conditional draw, including when the list holds almost all reference mass.

For q free bits, the tree takes $O(|A|q)$ preparation and storage after coordinate conversion; conversion can cost $O(|A|q^2)$. Traversal costs $O(q)$ plus the inverse map. The positive recursion also avoids subtracting nearly equal masses. Very small probabilities still require stable arithmetic.

5 Generate states by testing reliable parity coordinates

The independent reference suggests a search rule as well as a sampler. Let F span every nonzero score character. Each retained parity disagrees with its favored sign with probability

$$p_j = \frac{1}{1 + e^{2|H_j|}}.$$

Enumerate assignments with at most t disagreements, then evaluate their complete physical scores. This is related to ordered-statistics decoding's reliable-basis reprocessing[2]. The number of tested assignments is

$$N_t = \sum_{j=0}^{\min(t,r_F)} \binom{r_F}{j} = O(r_F^t) \quad \text{for fixed } t. \tag{18}$$

It supplies a finite alternative to SCL discovery. Increasing the radius changes the polynomial degree.

If $r_F = q$, an assignment identifies one payload. Otherwise it identifies 2^{q-r_F} equally weighted payloads, since all score characters lie in the span. Use that multiplicity in its mass and sample the free bits uniformly. A parity cell cannot be treated as one full message in the outer count decoder.

The reference flip count is a sum of independent Bernoulli variables. A suffix dynamic program capped at $t+1$ computes $\tau_t = \mathbb{P}_Q(\text{more than } t \text{ flips})$ in $O(r_F(t+1))$ work and storage. The same tables sample exactly conditional on exceeding t. Thus the refined envelope $U_t = Z_{A_t} + U\tau_t$ has both a computable total and an accessible complement sampler. An SCL list can instead supply A, or be combined with this search after deduplication.

5.1 One explored cell repairs the triangle

For $s = a(x + u - xu)$, retain reference $ax + au$ and explore only its favored cell $(x, u) = (1, 1)$. Replacing that cell's envelope weight e^{3a} by its exact weight e^a gives

$$U_0 = 3e^a + e^{-a}, \qquad Z = 3e^a + e^{-3a}, \qquad Z/U_0 \longrightarrow 1. \tag{19}$$

The single-flip cells already attain their envelope weights, so exploring them does not improve this bound. At $a = 8$, acceptance changes from 3.38×10^{-7} to .999999962.

The explored cell nevertheless holds only asymptotic target mass $1/3$. Its omitted-mass bound tends to $2/3$. Restricting inference to that cell would be poor; the refined full-space proposal succeeds because its direct complement sampler also visits the other modes. Improving a proposal and retaining enough states for list-only inference are different objectives.

5.2 Repeated conflicts require a different representation

For m independent triangles, target flip count tends to Binomial$(m, 2/3)$ as a grows. A fixed-radius list loses target mass as m increases. More sharply, for fixed m and $t < m - 1$,

$$\tau_t \sim \binom{2m}{t+1} e^{-2a(t+1)}, \qquad U \asymp e^{3ma}, \qquad Z \asymp 3^m e^{ma},$$

so refined acceptance still decays proportionally to $e^{-2a(m-t-1)}$. At $a = 4$, radius-zero acceptance is .999888 for one triangle but 3.82×10^{-10} for four. Radius one only raises the latter to 3.25×10^{-7}. A growing search radius can therefore recover exponential work.

These independent triangles have a cheaper exact solution: evaluate each four-state component separately. In general, express characters in the retained basis and connect coordinates appearing in a common factor. If component ranks are r_i, partitions and samples factor exactly, with enumeration costing roughly $\sum_i D_i 2^{r_i}$ after constructing the coordinate maps. Small joint groups can also replace parts of a reference, while omitted factors retain explicit envelope bounds. More general elimination costs depend on joint separator width, not the rank of each individual factor.

This is an exact control for the counterexample, not evidence of small components in the selected code. With every nonzero generator character present, the checked $(b, d) = (6, 32), (32, 128), (88, 512)$ constructions (CRC8, CRC16 and CRC16 respectively) each have a single component of rank b. Residual-specific zero coefficients, conditioning or an explicitly bounded factor approximation may simplify them; useful small decompositions remain to be established.

6 Reuse calculations across nearby residuals

Repeated proposal construction may dominate inference. For the same encoder, noise and allowed words, compare residuals r_0 and $r = r_0 + \Delta r$. Their exact score difference is

$$\Delta s(w) = \Delta r^T \phi(w)/\sigma^2.$$

Choose bounds $l \le \Delta s(w) \le u$ and write $\Omega = u - l$. Unit energy gives the interval $[-\|\Delta r\|/\sigma^2, \|\Delta r\|/\sigma^2]$ without searching words. Grouping the difference score by pattern gives a possible refinement:

$$l = \min_p \left(c_{p,0} - \sum_{g \neq 0} |c_{p,g}| \right), \qquad u = \max_p \left(c_{p,0} + \sum_{g \neq 0} |c_{p,g}| \right).$$

Intersect the two intervals. With cached row groups, the second calculation costs $O(Pd)$.

6.1 Transfer conditional accuracy bounds

Since $\pi_r \propto \pi_{r_0} e^{\Delta s}$,

$$\mathrm{TV}(\pi_r, \pi_{r_0}) \le \tanh(\Omega/4). \tag{20}$$

For $\Omega = 0$ the laws agree. Otherwise, let $X = e^{\Delta s} \in [a_-, a_+]$ and $m = \mathbb{E}X$. Convexity bounds the variation by $(a_+ - m)(m - a_-)/[(a_+ - a_-)m]$. Maximizing at $m = \sqrt{a_- a_+}$ gives the displayed bound. A cached approximate law with error η_0 has reused error at most $\min\{1, \eta_0 + \tanh(\Omega/4)\}$.

If a list has certified omitted mass at most ε_0 at r_0, then

$$\pi_r(C^c) \le \frac{e^{\Omega}\varepsilon_0}{1 - \varepsilon_0 + e^{\Omega}\varepsilon_0}. \tag{21}$$

Outside weights increase by at most e^u while inside weights decrease by at most e^l; comparing their odds proves the result. A desired tolerance $\varepsilon_\star > \varepsilon_0 > 0$ permits reuse if

$$\Omega \le \log \frac{\varepsilon_\star(1 - \varepsilon_0)}{\varepsilon_0(1 - \varepsilon_\star)}.$$

This can avoid another polar search. It certifies one conditional list, not complete multiuser coverage.

6.2 Reuse a proposal with exact correction

If a cached proposal satisfies $e^{s_{r_0}(w)} \le U_0 q_0(w)$, then $e^{s_r(w)} \le e^u U_0 q_0(w)$. Draw from q_0 and accept with $e^{s_r(w)}/(e^u U_0 q_0(w))$. The new conditional is exact, and an available lower bound $Z_0^- \le Z_{r_0}$ gives

$$\mathcal{A}_r = Z_r/(e^u U_0) \ge e^{-\Omega} Z_0^-/U_0. \tag{22}$$

Thus the explored-state refinement and residual reuse combine through the same pointwise envelope. Rejection can again be capped with a hold on failure, or q_0 can be used in MH.

For a fixed context, $C_0 = U_0/Z_0^-$ gives $\pi_r/q_0 \le e^{\Omega} C_0$. The independence-MH transition therefore dominates $e^{-\Omega}\pi_r/C_0$, giving contraction at most $(1 - e^{-\Omega}/C_0)^t$ after t moves. Indeed its off-diagonal mass is $\min\{q_0(w), \pi_r(w)q_0(v)/\pi_r(v)\}$ and each term has this lower bound. The bound does not establish mixing of the joint chain whose outside users keep changing.

6.3 Account for the cache itself

A fixed bank of centers can be prepared from y. Select a center using the outside residual, so a single-slot move uses the same proposal forward and backward. Pair moves require the corresponding reverse choices. History-dependent adaptation needs a separate validity argument.

With J_0 centers and J queries, charge J_0 constructions plus selection, comparison and physical rescoring for all J queries. Naive center selection costs $O(J_0 n)$ per query; grouped comparison adds $O(Pd)$. Low noise magnifies residual differences by $1/\sigma^2$, potentially requiring many centers. These bounds identify when reuse is safe, not whether useful residuals admit a small cache.

7 Discover a correlated pair without scanning payload pairs

Single-slot proposals can miss configurations that require two simultaneous replacements. A bounded pair search is therefore worth testing. Fix the other $K-2$ hypothesized slots and let r be their subtracted residual. First consider the hypothesis that both proposed messages use pattern p. Report 7 already gives full-alphabet pair contractions and a noiseless same-pattern identifiability criterion. The question here is different: can a finite noisy search propose useful pairs without enumerating 2^{2b} payload pairs?

7.1 Use a four-symbol polar channel

Put $a_i = \sqrt{e_{p,i}}$ and $r_i = r_{r(p,i)}$. For coded bits $x, z \in \mathrm{GF}(2)$, the local channel weight is

$$W_{p,i}(x,z) = \exp\left[-\frac{(r_i - a_i[(-1)^x + (-1)^z])^2}{2\sigma^2}\right]. \tag{23}$$

This is exact for the fixed outside context and same-pattern hypothesis. Because both users employ the same binary polar transform, their paired input symbols form an additive $\mathrm{GF}(2)^2$ transform. For two four-entry channel arrays A, B, its kernel $(u,v) \mapsto (u \oplus v, v)$ gives

$$W^-(u) = \tfrac{1}{4} \sum_{v \in \mathrm{GF}(2)^2} A(u \oplus v) B(v), \qquad W^+(v;u) \;\propto\; A(u \oplus v) B(v). \tag{24}$$

Joint polar decoding on multiple-access channels is established; see Şaşoğlu, Telatar and Yeh and Nasser and Telatar. The proposed use here keeps the existing encoder, frozen positions and two CRC constraints.

A deterministic list search extends a path by four symbols at an information position and by $(0,0)$ at a jointly frozen position, retains at most L_2 paths, and checks both CRCs. Its principal arithmetic can be $O(16 L_2 d \log d)$, with $O(4L_2 d)$ working memory using a standard shared-path implementation; sorting all extensions adds $O(4L_2 d \log(4L_2))$. Re-encode and physically rescore surviving pairs. This last step also restores the unoccupied-coordinate likelihood terms needed to compare patterns.

The local kernel identities are exact; a finite SCL list is an approximation. Its usual recursion marginalizes free future symbols, without fully imposing future frozen/CRC constraints, and pruning can discard the useful pair. It yields neither the constrained pair partition nor an exact posterior draw. The existing frozen schedule was not designed here as a new two-user MAC code, so the cited coding theorems do not establish its finite-list performance.

7.2 A cheaper XOR guide has a missing code factor

There is also a route through two ordinary binary polar searches. Write the ordered payload pair as $(v, v \oplus \delta)$, and define

$$m_i(\delta) = \mathbf{1}\{(L\delta)_i = 0\}, \qquad \theta_i = 2a_i r_i/\sigma^2, \qquad \kappa_i = 2a_i^2/\sigma^2.$$

The signals cancel where $m_i = 0$ and add where $m_i = 1$. After removing the common factor $e^{-\|r\|^2/(2\sigma^2)}$, the exact pair weight is

$$f_p(v,\delta) = \exp\left[\sum_i m_i(\delta)\{\theta_i \chi_{\ell_i}(v) - \kappa_i\}\right]. \tag{25}$$

For fixed δ, this is an ordinary polar likelihood in v, with LLRs $2m_i(\delta)\theta_i$. Under the IID uniform source conditioned on p, v and δ are independent uniform payloads; $\delta = 0$ correctly represents repeated messages.

Marginalizing each sign independently suggests XOR-code LLRs $\gamma_i = \log\cosh\theta_i - \kappa_i$. But the resulting weight G_p omits code consistency:

$$G_p(\delta) = \exp\left[\sum_i m_i(\delta)\gamma_i\right], \qquad \sum_v f_p(v,\delta) = 2^b G_p(\delta)\mathcal{C}_p(\delta), \tag{26}$$

$$\mathcal{C}_p(\delta) = 2^{-b} \sum_v \prod_{i:m_i(\delta)=1} \frac{e^{\theta_i \chi_{\ell_i}(v)}}{\cosh\theta_i}.$$

$\mathcal{C}_p = 1$ when all active rows are distinct, nonzero and linearly independent. Constant, repeated or dependent rows generally prevent that simplification. $\mathcal{C}_p$ is a likelihood correction, not a probability. On one tiny $b = 2, d = 8$, CRC4 fixture, the normalized guide probabilities for $\delta = 00, 01, 10, 11$ are $(.508, .399, .058, .035)$, whereas the exact probabilities are $(.203, .765, .014, .019)$: the preferred difference reverses.

One bounded candidate procedure decodes a list of L_δ differences using γ, then a conditional list of L_v payloads for each difference, and physically rescores the resulting pairs. Ignoring list sorting, its per-pattern search cost is $O((L_\delta + L_\delta L_v) d \log d)$. The first list can lose a useful difference before the exact score is evaluated. The earlier character envelopes can instead bound the masked partition in (26), but constructing that bound costs an inner operation for each tested difference. Neither route is selected as the winner. Report 7's existing rank condition for noiseless unordered-pair identification does not resolve this noisy discovery problem.

7.3 Normalize the finite proposal and retain the count target

For a fixed residual, pool and deduplicate the proposed *ordered* full-message pairs in $C(r)$. With $g_r(u_1, u_2) = \exp[-\|r - \phi(u_1) - \phi(u_2)\|^2/(2\sigma^2)]$, define

$$q_C(\mathbf{u} \mid r) = \frac{\mathbf{1}\{\mathbf{u} \in C(r)\} g_r(\mathbf{u})}{\sum_{\mathbf{z} \in C(r)} g_r(\mathbf{z})}, \qquad q(\mathbf{u} \mid r) = (1 - \alpha) q_C(\mathbf{u} \mid r) + \alpha / M^2. \tag{27}$$

Use the uniform pair law if the list is empty. Its draw requires only random message bits, and its normalization requires only the finite list. With fixed or uniformly selected slot indices, the IID ordered-slot prior cancels, and the block MH acceptance is

$$\min\left\{1, \frac{g_r(\mathbf{u})}{g_r(\mathbf{w})} \frac{q(\mathbf{w} \mid r)}{q(\mathbf{u} \mid r)}\right\}. \tag{28}$$

The outside residual is identical in both directions. A stochastic factorized same-pattern proposal instead has density $q_P(p \mid r) q_\delta(v \oplus u \mid p, r) q_v(v \mid v \oplus u, p, r)$ for the ordered pair $((p, v), (p, u))$; its actual forward and reverse densities must be evaluated.

If an unordered pair list is used, divide each pair's proposal mass equally among its two orderings, or its single ordering for a repeated message. There is no additional factorial factor in the ordered-slot update: projecting all K slots to counts already produces the IID count prior. Choosing slot pairs according to their current messages requires the corresponding reverse selection probability. Distinct-message sampling requires its own exclusions and proposal normalization.

For J_2 residual contexts and R_p tested patterns per context, charge the pair-search cost $J_2 R_p$ times, physical rescoring, and any $O(Pd)$ pattern screening in every context. The bank still costs $\Theta(Pd)$ memory; a retained pool contains up to $R_p L_2$ ordered pairs in addition to the per-pattern working memory. A naive scan of possible slot pairs adds $O(K^2)$ choices. For an IID prior pair, the probability of a common pattern is only $1/P$. With different patterns, overlaps can couple different polar symbol indices, so (24) is generally no longer an aligned memoryless channel. This proposal therefore addresses a restricted correlated move; full support alone gives no useful discovery or mixing bound for the remaining configurations. Count-tail estimation and the PUPE readout are still required after these moves.

Finite verification and next boundary. A separate identity script checks all 16 payload pairs in each of 20 tiny residual fixtures, the four-symbol kernel, and full-support pair MH detailed balance on 64 ordered states. It implements no joint SCL decoder and supplies no performance result. A useful next experiment is a bounded conditional comparison of joint four-symbol and XOR-first candidate recall, cost, and CRC-valid list size at growing b, d, followed by tests in estimated multiuser residuals. Those measurements would decide whether this operation merits an end-to-end bridge.

8 Keep the final count calculation sparse

For H weighted legal configurations $a^{(h)}$ with normalized weights ρ_h, their measure has tails

$$\widehat{T}_{w,j} = \sum_h \rho_h \mathbf{1}\{a_w^{(h)} \geq j\}.$$

Accumulate a histogram of positive counts for each observed word and take reverse cumulative sums. Since $\sum_w \max_h a_w^{(h)} \le \sum_{h,w} a_w^{(h)} = HK$, storage for nonzero tail levels is $O(HK)$, not $O(MK)$. A heap of next tails allocates the output in $O(C_s + K \log C_s)$ work, where $C_s \le HK$ is the number of observed words. Labels, waveform storage and inference remain separate costs.

For a retained multiset beam, weights are proportional to the physical likelihood divided by $\prod_w a_w!$, normalized over unique retained multisets. This is the posterior conditioned on the beam. Ordered-slot samples instead use prior M^{-K}; adding another factorial factor would count the source twice.

8.1 Restricting outputs differs from restricting explanations

For a candidate set C, define

$$\mu_{\text{out}} = \frac{1}{K}\mathbb{E}\sum_{w\notin C} a_w, \qquad \varepsilon_C = \mathbb{P}\left(\sum_{w\notin C} a_w > 0 \mid y\right),$$

where the expectation is under the full posterior. Every C-supported action has risk at least μ_{out}. If its tails are still computed under the full posterior, the best such action has *excess* risk at most μ_{out}: remove outside entries from a global optimal action and refill inside. Uniform retained-tail error η adds at most 2η. Refilling requires nonempty C for IID counts, or $|C| \ge K$ for distinct counts.

Conditioning the entire posterior on all words belonging to C gives a different bound. Its Bayes action has excess risk at most ε_C, because the full law is a mixture of the conditional law and its complement. Conditional inference regret δ_C gives $(1-\varepsilon_C)\delta_C + \varepsilon_C$.

For example, assign probability $1/4$ to a^4 and $3/4$ to b^3c, with $C = \{a, b\}$. The outside-copy fraction is $3/16$, but conditioning selects a^4, with excess risk $1/2$. The full-posterior constrained action is b^3a. Their risks are .75 and .375, versus Bayes risk .25. This decision-law example shows why candidate recall alone cannot justify conditioning the posterior.

8.2 Representation can be small even when obtaining it is hard

If H independent exact posterior samples were available, Bernoulli concentration and a union bound give

$$\mathbb{P}\left(\max_{w,j} |\widehat{T}_{w,j} - T_{w,j}| > \eta\right) \le 2MKe^{-2H\eta^2}. \tag{29}$$

Thus $H \ge [B\log 2 + \log(2K/\delta)]/(2\eta^2)$ suffices for excess risk at most 2η with probability $1-\delta$. Unseen words have empirical tail zero, so sparse readout suffices despite the all-word bound. Only $\log M$ enters this sample count. Producing the independent samples is the unresolved step; correlated chain states or weighted importance draws do not satisfy this premise automatically.

9 Use overlap parities to separate a mixed-pattern block

The same-pattern channel does not handle general mixed-pattern overlaps. Report 7 supplies their exact parity factorization, but its pair operator still groups and copies all 2^b payload weights. Combining those parities with affine one-user envelopes gives another finite operation. Its cost is exponential in a chosen interaction rank; a small rank is a condition to test, not an assumption.

9.1 Condition one endpoint of every retained interaction

Fix m slot indices, their hypothesized ordered patterns $p_1, \dots, p_m$, and the residual r after subtracting the other slots. Removing the common residual norm and m unit self-energies leaves

$$S(v_1, \dots, v_m) = \sum_k s_k(v_k) - \sum_{k<l}\sum_{g,h} J_{kl}(g,h)\chi_g(v_k)\chi_h(v_l), \qquad s_k(v) = r^T\phi(p_k, v)/\sigma^2, \tag{30}$$

where $J_{kl}(g,h)$ sums $\sqrt{e_{p_k,i}e_{p_l,j}}/\sigma^2$ over shared resources whose local generator rows are g,h. Terms with a zero row are constants or unaries; absorb them first. This can reduce the interaction rank without approximation.

For slot k, choose a character subspace W_k with independent row basis H_k, rank r_k. Call a remaining factor *covered* when $g \in W_k$ or $h \in W_l$. Condition $\alpha_k = H_k v_k$. For $g \in W_k$, its sign is now a known function of α_k. A covered factor therefore becomes a constant if both endpoints are fixed, or shifts the unary coefficient at its unfixed endpoint. Keeping all covered factors gives exactly

$$S_W = c(\alpha) + \sum_k t_{k,\alpha}(v_k), \qquad H_k v_k = \alpha_k,$$
$$Z_W = \sum_\alpha e^{c(\alpha)} \prod_{k=1}^{m} Z_{k,\alpha}, \qquad Z_{k,\alpha} = \sum_{v: H_k v = \alpha_k} e^{t_{k,\alpha}(v)}. \tag{31}$$

There are 2^R sectors, $R = \sum_k r_k$. Every payload tuple belongs to exactly one, so no extra 2^{-R} factor appears. Write each affine support as $v = v_0 + Nz$ and use the restricted character algebra. It retains all polar and CRC constraints; ordinary unconstrained SCL does not automatically enforce these additional parity conditions.

For a pair, let W_p span all nonconstant left overlap rows and set $W_q = \{0\}$. Conditional on the left overlap signs, the right law is an ordinary polar likelihood with shifted LLRs; the left law is affine-restricted. This needs 2^{r_p} sectors, versus $2^{r_p+r_q}$ entries in the two-sided overlap table, and avoids its full-payload grouping stage. Reversing the orientation gives another choice. Neither sector's polar partition is free.

For several users, form each slot's *joint* row span across its assigned interactions. Rows shared by different neighbours are charged once. A three-user star control below has two rank-two edge interfaces at the center but only rank two jointly, hence four sectors. Independent edge summaries would fail to enforce that they came from the same central payload.

9.2 Limit rank, bound the omitted factors, or correct exactly

With a rank budget, some factors may remain uncovered. Set

$$D = S - S_W, \qquad T(W) = \sum_{\text{uncovered } (k,l,g,h)} |J_{kl}(g,h)|.$$

Then $|D| \le T$, giving

$$e^{-T} Z_W \le Z \le e^{T} Z_W, \qquad \mathrm{TV}(\pi, \pi_W) \le \tanh(T/2). \tag{32}$$

The variation bound uses log-weight oscillation at most $2T$. It bounds every conditional count-tail error. For this fixed outside configuration and pattern tuple, a Bayes action under π_W has conditional PUPE regret at most $2(m/K)\tanh(T/2)$: only m true copies can change, so each action's loss varies by at most m/K. This does not bound the stationary error of approximate block updates whose contexts keep changing.

Exact correction need not evaluate the inner partitions. For every sector, construct available pointwise envelopes $e^{t_{k,\alpha}} \le U_{k,\alpha} q_{k,\alpha}$ and lower partition bounds $L_{k,\alpha}$ using Section 3 or its refinements. Define

$$U_W = \sum_\alpha e^{c(\alpha)} \prod_k U_{k,\alpha}, \quad L_W = \sum_\alpha e^{c(\alpha)} \prod_k L_{k,\alpha}, \quad Q(\mathbf{v}) = \frac{e^{c(\alpha)} \prod_k U_{k,\alpha}}{U_W} \prod_k q_{k,\alpha}(v_k). \tag{33}$$

Sample a sector with the displayed finite weights, then sample its independent affine payload laws. The sector is determined by the resulting tuple, so its density is evaluable. The full physical weight satisfies $e^S \le e^T U_W Q$. Accept with $e^S/(e^T U_W Q)$ to obtain an exact fixed-pattern block draw, with

$$\mathcal{A} = Z/(e^T U_W) \ge e^{-2T} L_W / U_W. \tag{34}$$

Alternatively use this full-support proposal in a valid block MH update. The IID prior is uniform on ordered slots, including repeated messages; count factorials must not be added again. Distinct sampling introduces further exclusion factors requiring separate treatment.

9.3 Choose and pay for the representation

An executable rank-limited heuristic starts with empty spans, adds an endpoint row that captures the largest total uncovered coefficient magnitude after taking its span, and stops at budget R. It is not the earlier optimal independent-basis selection. Even one-sided selection can fail: give keys $e_1, e_2, e_1 + e_2$ weight 9 each and keys e_3, e_4 weight 10 each. With rank budget two, greedy captures 20; the first two coordinates capture 27. General cover selection includes graph vertex-cover choices when every slot has one bit. Small families of declared candidate covers can therefore be compared without claiming a globally optimal choice. Naively testing all endpoint rows and rescanning factors after each addition costs $O(RE^2b^2)$ binary/scalar work for E factors; the complete one-sided pair cover avoids that search.

Let E count grouped cross-factors and $C_{k,\alpha}$ be the cost of constructing a restricted unary envelope. After geometry and binary-linear preprocessing, setup costs

$$O\left(2^R\left[E + \sum_k C_{k,\alpha}\right]\right)$$

with the worst sector cost understood. Independent-basis envelopes have polynomial construction cost in d, b, but their acceptance can be exponentially small. Store 2^R sector weights and rebuild the selected sector, or cache its inner proposals at additional memory cost. Rebuilding pays the selected inner construction cost on every attempt. Charge physical rescoring and all repeated attempts; the certified mean-attempt bound is $e^{2T}U_W/L_W$. A larger cover can simplify inner laws enough to compensate for its extra sectors, so minimum rank alone is not the runtime objective.

For J contexts and H tested pattern tuples per context, pay these costs JH times unless a specific reuse bound applies. Exhaustive pattern search costs P^m tuples; conditioning on one tuple does not discover it. Geometry may be cached, but residual-dependent unary coefficients change. The pattern bank still costs $\Theta(Pd)$, and complete inference still needs count-tail readout. For independent uniform d-subsets of n resources, expected pair overlap is d^2/n; increasing payload length does not ensure small overlap rank. Rank b still gives 2^b sectors in this one-sided procedure. Low noise also amplifies T by $1/\sigma^2$.

9.4 Evidence and relationship to earlier work

Conditioning variables to simplify inference is established; see Bidyuk and Dechter, *Cutset Sampling for Bayesian Networks*. Report 10's DEC-09 already admits conditioned feature cutsets. The concrete extension here combines Report 7's physical overlap characters, Report 9's affine inner algebra and Report 11's pointwise envelopes to avoid an explicit payload table in a fixed-pattern block operation. This targeted comparison does not establish research novelty.

The independent script `tests/polar_tractability_structural_checks.py` checks all 512 payload tuples in each of 12 tiny three-user fixtures, four cover choices, a same-pattern duplicate control, partition bounds, normalized proposal correction and the greedy counterexample. Maximum discrepancy is 5.33×10^{-15}. In the first fixture the exact rank-two cover uses four sectors; its envelope acceptance is .350, illustrating that exact separation still leaves inner sampling cost.

A placement/generator census for $B = 12, b = 6, d = 32, n = 256$, seed 3201, gives 31,941 one-sided sectors over all 2,016 distinct unordered pattern pairs, versus 1,093,347 two-sided nonconstant table entries. Constant-row interactions were absorbed into unaries in both counts. These are representation counts, not a runtime or large-payload result: sectors require inner calculations. A measured rank/acceptance/construction-cost comparison on actual residuals is the next useful boundary.

10 Combine the operations and charge the complete cost

Let J count residual contexts, R_p patterns decoded per context, L_s the SCL width, C pooled words, W beam width and H complete proposals. An efficient SCL implementation costs $O(L_s d \log d)$ time and $O(L_s d)$ space[1]; prototype copying, sorting and CRC work require their own accounting.

Operation	Principal work	Remaining question
Polar lists then multiset beam	$J[Pd + P\log P + R_pL_sd\log d]$; then $KWCd$ plus bookkeeping	Do the lists and retained configurations cover useful explanations?
Parity reference	Per pattern/context: $Dq^2 + q^3 + D\log D$; draw: $q^2 + Dq$	Is its envelope tight enough to justify construction?
Radius-t refinement	N_tDr_F scores and $r_F(t+1)$ suffix tables	Is a small radius effective on the residuals encountered?
Arbitrary-list refinement	Coordinate conversion, exact scores and $\lvert A\rvert q$ tree work	Does reduced envelope slack repay the preparation cost?
Residual reuse	Center selection, comparison and new physical scores	Can enough contexts share a small bank?
Joint same-pattern list	$16J_2R_pL_2d\log d$ plus sorting, screening and scores	Is joint discovery useful often enough to repay its cost?

Here D is the number of grouped characters. Bounds use elementary arithmetic; binary packing and cached maps can improve implementation costs. All-pattern preparation multiplies by P. Persistent encoder storage is $O(Pd + dq)$, cached basis maps can add $O(Pq^2)$, and dense residual storage costs $O(n)$ per retained context. Dense Gaussian guidance additionally requires covariance work, up to $O(n^3)$ factorization and $O(n^2)$ memory.

Fresh discovery at every beam parent gives J of order KW; sequential importance sampling can give $J = HK$. Sampling H ordered tuples from their actual proposal $q(W \mid y)$ uses weights

$$\omega(W) = \frac{M^{-K}\exp[-\|y - \sum_k \phi(w_k)\|^2/(2\sigma^2)]}{q(W \mid y)}.$$

A whole-tuple prior mixture bounds these unnormalized weights by $1/\alpha$; separate stagewise mixtures only give α^{-K}. Neither controls evidence-relative variance by itself. Similarly, good one-slot rejection acceptance does not guarantee global mixing. Shared lists, small exact factor groups, selective refinement and residual reuse can combine, but their joint construction and discovery costs must be counted.

11 Finite decoder implementations

The following IID-source variants now use procedural waveforms and sparse count tails. All know K and the physical noise variance. They are complete finite-budget receivers, not exact posterior solvers. Their numbering continues Report 10's bank and the DEC-14–24 combination study.

ID	Discovery or proposal	Inference and count decision
DEC-25	Screen patterns; ordinary polar SCL lists	Bounded multiset beam; exact posterior restricted to retained states
DEC-26	Screen patterns; reliable-basis flip candidates	Same multiset beam and restricted count tails
DEC-27	Independent-parity envelope across all patterns	Physically corrected single-slot MH; empirical count tails
DEC-28	Replace selected envelope weights by exact weights	Same MH kernel with direct complement sampling
DEC-29	Freeze a small bank of repaired proposals	Select by outside residual; exact physical MH correction
DEC-30	Four-symbol same-pattern joint polar lists	Alternate pair-list MH and single-slot SCL-list MH
DEC-31	XOR-first same-pattern pair lists	Same alternating kernels; exact retained-pair scoring
DEC-32	Single-slot polar list mixed with the uniform prior	Corrected single-slot MH; empirical count tails

The beams merge identical partial multisets. Their continuation score uses the isotropic residual guide; at the final depth, each retained multiset has weight proportional to $[K!/\prod_w a_w!]\exp[-\|y - \Phi a\|^2/(2\sigma^2)]$. The chains start from a bounded SCL beam, update ordered slots, and add no further factorial term. Pair moves have full support through their uniform ordered-pair mixture; ordinary single-slot moves also change patterns. Same-pattern list proposals therefore do not impose a same-pattern posterior restriction. Each finite chain has initialization and sampling error; invariant kernels alone do not bound that error.

Refinement explores a capped set of actual payloads; the exclusion tree retains every other payload. A cache is fixed after initialization, and its selected proposal depends only on the outside residual, which is

identical forward and reverse for that block. Each proposal's actual normalized density enters MH. No stale likelihood or uncorrected adaptive chain is substituted. An empty CRC-valid discovery list remains empty: list proposals then use the prior, while a beam parent receives one declared prior draw. The latter prevents an empty output but supplies no candidate-coverage guarantee.

The procedural interface stores integer message IDs, on-demand waveforms, retained configurations and only visited-word tail entries. It has no M-length array. The implementation currently caches a dense n-vector per generated waveform, costing $O(nC_{\text{generated}})$ within a call; budgets still control this cost. Parity proposals rebuild all P pattern envelopes unless cached. Pair recursion uses explicit array copies, so an optimized SCL complexity claim would be premature. The common small-B harness retains its explicit encoder and converts sparse hard counts to its legacy array format. That harness is not itself a large-B execution path.

Independent tests recover REF tails from exhaustive tiny beams, verify repaired draw laws and single/pair MH balance, and match procedural outputs to the legacy adapter. A separate bounded $B = 100$ test uses $P = 4, b = 98, d = 128, n = 256$, CRC16 and integer IDs beyond 64 bits. All variants execute for $K = 1, 2$ without an alphabet table; the noiseless single-user observation is recovered at declared positive variance. This is an execution/control test, not evidence of useful multiuser PUPE at $B = 100$. The new mixed-pattern parity-cover operation in Section 9 remains a checked algebraic construction; it is not silently included in DEC-25–32.

12 Established results and remaining questions

Normalized proposals, finite-set repair, residual reuse and sparse count decisions avoid payload enumeration in declared operations. Overlap conditioning extends these to mixed patterns, but interaction rank and inner acceptance can still be costly. The counterexamples rule out automatic sufficiency of independent factors, fixed flip radii or small lists.

Next, compare corrections from matched initial configurations and increase payload length at declared P, n, d, K scaling. Measure parity-cover rank, setup and acceptance on actual residuals. The present identities and smoke checks establish no fixed-PUPE large-payload scaling result.

Relation to existing methods. Reliable-basis reprocessing and polar list decoding are established[2, 1], as are combinations of search, bounding and importance sampling, and partition refinement for rejection sampling[3, 4]. Approximate codebook probability has also been used for polar reliability and soft outputs[5]. The present operations specialize these principles to the balanced ODMA posterior and count decision. A research contribution must identify a specific new construction or guarantee, or a useful measured improvement in complete-receiver cost and PUPE.

Verification. The four earlier algebra scripts check 40 character fixtures, 30 refinement fixtures, 40 balanced residual pairs and 20 same-pattern pair fixtures, including the stated failure controls, with discrepancies below 10^{-11}. The mixed-pattern checks and independent implementation tests are described above. Commands, finite budgets and local smoke outcomes are indexed in the experiment bank and research log. These double-precision checks are not interval certificates or evidence of global mixing.

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